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**Research and Theory as Necessary Tools for
Organizational Training and Performance Improvement Practitioners**

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Abstract

Human Performance Technology (HPT) is the applied study and practice of improving organizational performance through training and non-training interventions. For practitioners working in this area that identify themselves as an HPT practitioner, organizational training and performance (OTP) specialist, or instructional designer—offering the right intervention set requires understanding of how humans work and function internally and within organizations that are bounded by environmental, societal, and economic realities. The HPT field is rooted in a multi-disciplinary knowledge base and has its own models and theories, which are generally developed by practitioners to guide their practice. Because HPT follows a systematic, data-driven process akin to action research, HPT practitioners are applied researchers even if they do not see themselves as such. What we need is more collaboration between scholars, practitioners, professional associations, and businesses to create a culture of engagement and involve everybody in the validation of applied research and theory development.

Keywords: Research, theory, practice, human performance technology, business, organizations

Going Back to the Roots of Organizational Training and Performance

As the mission statement for the Association for Educational Communications and Technology (AECT) Division of Organizational Training and Performance (OTP) indicates, OTP's aim is to create and nurture a community whose members are concerned with improving human performance in organizations through theory and research driven training and performance improvement interventions. The OTP division in AECT is not alone in this endeavor. Other professional organizations such as the International Society for Performance Improvement (ISPI) and the Association for Talent Development (ATD) aspire to similar goals. And similar to AECT's OTP division, ISPI and ATD have gone through some evolutions and name changes to evolve into what they are today. We believe that understanding this history and common ground between these organizations, even at the surface level, will help us to situate where theory and research stand for OTP practice. AECT's OTP was known as Training and Performance (TP) until late 2014. TP was the outcome of a merger in 2000 of the former Industrial Training and Education Division and the Division for Learning and Performance Environments. Similarly, when ISPI was first founded, it was known as National Society for Programmed Instruction (NSPI), which later became National Society for Performance and Instruction, preserving the same acronym of NSPI, until it became today's ISPI. ATD for many decades was known as the American Society for Training and Development (ASTD). The common thread for all three organizations is the focus on training, which underlines how the educational and instructional technology fields emerged: high demand to train masses of adults during and after World War II. With the introduction of general systems theory, a shift started to occur to supplement training with related interventions as scholars and practitioners realized that

a lack of skill or knowledge alone is not always the root cause of human performance problems in organizations or that training alone is not sufficient to address performance issues and problems.

As a result, a field called Human Performance Technology (HPT) evolved. HPT is the “study and ethical practice of improving productivity in organizations by designing and developing effective interventions that are results-oriented, comprehensive, and systemic” (Pershing, 2006, p. 6). HPT, today, is also referred to as performance improvement (PI), human performance improvement (HPI), workplace learning and performance improvement (WLPI), or organizational training and performance (OTP). For the sake of simplicity, we will present all of these terms as HPT for the remainder of this article because “*human performance technology*” better explains what practitioners of this field strive to do: to improve human performance by applying scientific and other accumulated knowledge systematically to performance issues. It also implies that HPT practitioners are not tasked or concerned to improve any performance; only the performance in organizations that involves humans. Furthermore, HPT is systemic and holistic in that it does not presuppose or favor training over other interventions.

In a similar special issue of TechTrends dedicated to workplace learning and performance improvement about 12 years ago, a group of scholars and authors highlighted the linkage between instructional design (ID) and HPT (Benson & Schaffer, 2004). Today, this connection still exists and it is more important than ever to recognize that instructional designers in organizational settings often find themselves working in teams and developing instructional and non-instructional solutions as part of a set of performance improvement interventions. In addition, issues related to education alone are rarely sufficient for performance improvement (Molenda & Pershing, 2004). Many of the founders of HPT were instructional designers and

today's HPT practitioners are mostly graduates of instructional design and technology programs. Experts in the field consider the skills and knowledge related to the performance improvement process as very important for the graduates of these programs, whether they become instructional designers or HPT consultants (Klein & Fox, 2004). Therefore, in our judgment all practitioners in instructional and performance technologies will find the ideas we discuss relevant to them and to their practice.

HPT as a Theory Driven Scientific Field

By definition technology is “the systematic application of scientific or other organized knowledge to practical tasks” (Galbraith, 1985, p. 11). HPT is called a technology because it is the systematic application of theories, models, and principles to address performance problems in organizations. Founders of HPT such as Thomas Gilbert, Joe Harless, Robert Mager, Dale Brethower, Donald Tosti, and Geary Rummler learned that there is power in observation, empirical and measurable evidence, and principles of reasoning (Dean & Ripley, 1997). They recognized that rigorous observations and measurements that are grounded in the scientific process are forceful means for improving individual and organizational performance. Further, they learned that the analyses of observations across performance improvement cases can lead to identification of relationships among variables that in turn lead to theory and model development. For fields of study such as HPT, which focus on applications for use in the everyday world, the knowledge domain yields validated principles, procedures, and tools for practice. For technology based fields, the knowledge base is anchored in applications of scientifically generated knowledge aimed at solving practical problems for a variety of societal institutions including government, industry, commerce, and so on. Given this logic chain, a *knowledge base* is foundational to human performance technology (HPT) as a field of study.

As organizations are open and dynamic systems that exist within economic realities, the field of HPT by necessity has a multidisciplinary theory base. Swanson (1999) proposed three theory bases as foundational to performance improvement theory: economic theory, systems theory, and psychological theory. He argues that:

[E]conomic theory is recognized as a primary to survive along with its financial metrics at the organizational level; systems theory recognizes purpose, pieces, and relationships that can enhance or strangle systems and subsystems; and psychological theory acknowledges human beings as brokers of productivity along with their cultural and behavioral nuances (p. 11).

In a similar vein, Dale Brethower noted that if knowledge, perspectives, and theories from fields such as economics, organizational development, business administration, labor relations, psychology, and accountancy help performance technologists to “... identify variables that support or interfere with excellent performance, it is knowledge we need” (Dean, 1977, p. 78). In our judgment, these scholars are presenting an eclectic view that we believe to be a truism for the field of HPT. We also believe that over time the list of supporting fields and disciplines has expanded. One that we personally value is research and inquiry given the tenet that performance improvement initiatives need to be grounded in actionable research using valid and reliable data and information.

In addition to informing HPT practices, theories also play a central role in developing performance improvement models. Theories provide guidance to practitioners when the theories are practical and have utility. Since the inception of the field, numerous HPT models have been created. These models can be classified under three main headings: diagnostic models, process models, and holistic models (Wilmoth, Prigmore, & Bray, 2010). Diagnostic models provide a

framework to analyze performance issues and they have been widely adopted by HPT practitioners to identify the actual need, cause, and gap for performance issues that are critical to the success of human performance improvement initiatives. Gilbert's (1978) behavioral engineering model, Harless' (1970) front-end analysis model, and Wile's (1996) synthesized performance model are commonly known and widely used diagnostic models. Process models help HPT practitioners examine issues beyond the performance problems, identify the potential interventions, and outline the process for addressing performance issues. Process models are generally linear, have grouped activities, seek out performance gaps, consider multiple interventions, and evaluate results with a feedback loop (Wilmoth, Prigmore, & Bray, 2010). ISPI's performance improvement HPT model (Van Tiem, Moseley, & Dessinger, 2012) and Pershing's HPT model (2006) are examples of often cited process models. Finally, holistic models are intended to represent how domains co-exist in a performance context. They typically depict domains in overlapping shapes; therefore, they are nonlinear unlike process models. Stock's (1996) three dimensional HPT model is an example of a holistic model.

Value Proposition for HPT and the Need for Research and Theory

For several decades now, organizations have been investing a great deal of their resources in their human capital. This has become more so with the transition from an industrial era to an information era, where processes and knowledge became more agile. The "State of the Industry Report," an annual research publication of the ATD, shows the magnitude of these capital investments: organizations in the US spend over \$1,200 per employee annually for learning (Miller, 2013). Given that training interventions are hardly a standalone solution to organizational performance issues, an annual spending of \$164 billion on training is just a

portion of what the US organizations actually spend in efforts to improve human performance. Performance improvement specialists and practitioners play a key role in determining where and how this budget is spent by making recommendations to their internal and external clients through a HPT process. Therefore, we HPT practitioners bear the responsibility to maximize the return on this investment by offering the best set of solutions and interventions and managing the associated change process. In order to achieve our worth, we need reliable and credible tools and techniques that are evidence based. Grounding our work in a research and theory base is critical for practitioners for two reasons. First, we need process models and theories that are proven to work. As an applied field, process models in HPT have not evolved from traditional research driven approaches; in most cases, they are developed by practitioners based on their experiences. However, to go beyond one's experience and earn utility for the field, all models need to be tested and validated. Such validation is essential for the value added aspect of HPT for organizations. Secondly, given that the nature of HPT practice concerns organizations and individuals, we HPT practitioners need to acquire at least a basic understanding of the social sciences that seek to understand and explain human nature, organizations, and the economy. Otherwise, interventions that we propose and develop will not be grounded in a commonly shared knowledge base making their value and utility free flowing and unworthy: that is, lacking in validity and reliability. This is akin to prescribing *snake oil* for the individuals and organizations that we serve.

Research: Maybe Not So Scary After All

As we have discussed, theories and models in HPT evolved from systematic practice. Theory and model development has not always been done by scholars and researchers; in many cases they were generated by practitioners. The word *research* because of its connotations may

sound intimidating to many people, not just HPT practitioners. Meriam-Webster Online Dictionary (n.d.) describes research as “1. careful or diligent search; 2. investigation or experimentation aimed at the discovery and interpretation of facts, revision of accepted theories or laws in the light of new facts, or practical application of such new or revised theories or laws.” As complicated as this may sound, many of us have most likely done research in our work or in our personal lives. If you are buying a new or used car, shopping for a baby crib for your unborn child, or as a performance consultant checking records of spending on learning in the past five years in a company, you are doing research. One of the key features of research is its systematic nature, which is tied to its rigor, also known as quality. To illustrate, when you are checking the safety ratings of a new car, you probably want to see the ratings from an independent institution such as National Highway Traffic Safety Administration (NHTSA) rather than a manufacturer’s own assessments. You may also want to see that these ratings are based on actual tests, they are not just hypothetical. Same is true when it comes to HPT practice. If you are contracted to an organization and handed a performance issue by your client stating that the sales department is underperforming and they need training on upselling, you may probably want to confirm first that the performance problem is actually due to the low sales and that sales people are to blame before proposing any solutions.

We HPT practitioners are action researchers. Although a universally accepted definition of action research is yet to be written, there are three key attributes of action research:

- It is oriented towards solving practical problems,
- Is participative in nature, and
- It utilizes systematic inquiry processes.

In a recent study, Duan (2014) examined published case studies related to HPT and found that HPT practitioners actually conform to the principles and processes of the applied research framework. The action research process framework involves a systematic and iterative cycle of data planning, data collection, and data analysis which is embedded in the overall HPT process, including elements such as needs analysis, the analyze, design, develop, implement and evaluate (ADDIE) instructional design process, or program evaluation. For example, needs or gap analysis involves collecting data from stakeholders such as clients, managers, and targeted employees through interviews, observations, or surveys; all of which are not different from conducting research. At the end of a training intervention, measuring the impact of training on employees using the Kirkpatrick Level 2 evaluation or the organization using the Kirkpatrick Level 3 evaluation requires collecting data again and observing differences in performance (Kirkpatrick & Kirkpatrick, 2006). Consequently, the nature of HPT initiatives requires that practitioners be researchers, even if practitioners do not see or call themselves as such.

A research perspective that requires more directive relates to the core outcome of the performance improvement process: intervention selection, design, development, and implementation. If a single intervention or intervention set does not close a performance gap, the HPT process fails. Therefore, performance improvement specialists and practitioners must work with intervention experts to identify and test interventions as to their relevance, applicability, and overall vitality and vigor via a rigorous research process that validates their utility. In short, interventions must be effective and efficient, produce results that are transferable, and produce positive returns on investments. This requires evaluative research that is a valuable component of the overall research and development process.

Finally, HPT scholars, specialists, and practitioners must be involved in conducting the research that is necessary to advance the HPT knowledge base through testing existing models, which leads to validation or refinement of these models and eventually theory development. To achieve this goal, we need collaboration among practitioners, academics that are researchers, and business managers given that this research cannot for the most part be conducted in laboratory like settings. However, conducting research in organizational settings is not simple and straightforward. One of the reasons that doing this research in organizations is challenging is that the research variables cannot be isolated or controlled like in laboratory settings. If permitted, such research needs to take place in natural or near natural work settings. As a result, other confounding factors can play into the outcomes. Secondly, HPT practitioners typically work under time and budget constraints. Completing a thorough analysis or evaluation that entails detailed data collection becomes difficult because stakeholders and clients may want to expedite the HPT interventions in order to respond to pressures from their environment. These three foci of HPT action research, intervention research, and foundational research are summarized in Table 1.

Table 1: Three Research Foci for the Field of Human Performance Technology

<i>How We Do It</i> Action Research	<i>What We Do</i> Intervention Validation Research	<i>Why It Is Meaningful</i> Foundational Research
Understand and improve the inquiry based processes and practices of the performance improvement enterprise.	Validate the selection, design, and development via testing of performance improvement interventions.	Grow the evolving knowledge base of HPT including its philosophy, history, foundations, models, assumptions, and dominant paradigms.

All in all, theory and research are inseparable from practices and developments in the HPT process. Swanson (1997) depicted this relationship as a theory—research—development—practice cycle, in which theory is at the center of the cycle and the linkage between each component is bidirectional (see Figure 1). In other words, he asserts that one can start the process at any point; for example from theory or from practice, but each component informs the other three. Similarly, other scholars in the field have emphasized the interdependence of practice and research. For example, Stolovitch advises practitioners to think like a researcher and researchers to think like a practitioner (2000). Recently, Pershing proclaimed his view that “... HPT is an intellectual enterprise and not a set of routine mechanistic principles and exercises” (Pershing, 2016, p. 37). He goes on to advocate for “... promoting an understanding of the history, foundations, and theories of HPT ... and explains that such understandings are central to dispelling an excess of myths, opinions, and propaganda ... that too often is marketed by mechanistic performance improvement specialists” (p.39). Given these perspectives, we believe that HPT foundations and theory are valid frameworks to evaluate performance improvement evidence and they are a viable means to gain insight to the variety of constraints and influences that impact HPT practices.

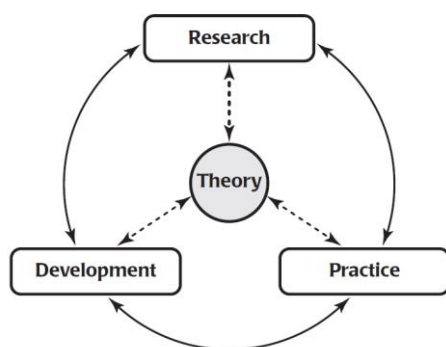


Figure 1. Theory-Research-Development-Practice Cycle. Reprinted from *Human Resource Development Research Handbook* (p. 14), R. A. Swanson & E. F. Holton III, 1997, San Francisco: Berrett-Koehler. Copyright 1997 by R. A. Swanson. Reprinted with permission.

Bridging the Gaps Between Theory,
Research, and Practice

There is constant demand for high performance in organizations (Swanson, 1999). Addressing the need for improved performance requires a systems approach and a theory-driven understanding of the elements that drive individual, team, and organizational performance. HPT as a practice-driven field has no shortage of performance models and tools, but most of these models are not theory-driven or validated by research. HPT practice without the backing of substantive theory and systematic research becomes isolated pockets of good practice at best that does not go beyond trial-and-error. Therefore, as Swanson emphasizes, maturation of HPT as a field and as a profession depends on systematically filling the void in theory and practice of performance improvement. We believe that this responsibility falls not only on HPT practitioners, but also on scholars in the field and academic and professional organizations as well as businesses.

We HPT practitioners can engage in research and theory in several ways. First, in order to truly improve organizational performance our methods for addressing performance issues in organizations should be evidence based and systematic from analysis thru evaluation. In other words, we need to have a solid understanding and working knowledge of the commonly used data collection and analysis methods in our field. Without properly collecting data on gaps and causes, processes, and outcomes, we cannot claim positive impactful change in organizational performance. In fact, both faculty of educational technology graduate programs and expert practitioners consider analysis, measurement, and evaluation skills more important than instructional technology skills, suggesting that graduates of the Instructional Design and Technology (IDT) programs should have these skills (Klein & Fox, 2004). Consequently, HPT

and instructional design practitioners should be systematic and must base their decisions on data for successful and value-added practice. This is the essence of applied or action research.

Second, we should follow research and theory closely in order to provide the best solution and add value to the organizations we serve. Adding value and creating sustainable change are foundational principles of HPT practice. Joining professional and academic organizations related to HPT such as AECT, ISPI, ATD, and the Academy of Human Resource Development (AHRD) would facilitate us to follow research as these organizations hold annual conferences to share research and best practices. Members of these organizations including academics, students, and practitioners publish books and articles in research journals, often sponsored or managed by these organizations. Moreover, these organizations are alternative venues to acquire the data collection and analysis skills we mentioned above as they offer workshops in an HPT/IDT context.

Third, we as practitioners should be more involved in generating research in the HPT field. Previous reviews of ISPI and AECT journals regarding HPT research reveal that the amount of empirical research is small and scope of the existing research is narrow (Conn & Gitonga, 2004; Klein, 2002; Marker, Huglin, & Johnsen, 2006). All of these reviews call for more research and propose research areas to close the gap in the literature. Klein (2010) summarizes these reviews and offers a short list of topics for future research. Similarly, in light of these reviews, expert opinions in the field (Huglin, Johnsen, & Marker, 2007), and our assessment of the field (authors, in press), below are some areas—rather than a complete list, of where practitioners can partake in future research:

- Intervention sets and processes – particularly focusing on noninstructional interventions

- Program evaluation – particularly focusing on evaluation for noninstructional interventions
- Validation of HPT models and tools

To publish and contribute to the HPT knowledge base, practitioners may either document their practice systematically as case studies or partner with HPT scholars to conduct and report more rigorous research. Publishing cases studies of HPT practice that applies models and principles known in the field, allows HPT scholars to validate these models and principles through analysis of the cases. As some practitioners may not possess the research skills to embark on more rigorous research in organizational settings, collaborating with HPT scholars can open the door to many possibilities for research. This partnership is also a win-win situation for HPT scholars because they do not always have easy access to organizations for conducting field studies. In either case, support from clients/organizations is critical for conducting and publishing research as they have a say in how much resource is allocated for data collection/analysis and how much information can be shared outside the organization.

Finally, engagement of graduate students in HPT practice and research early in their academic programs is important, regardless of whether they aspire to become practitioners or researchers. Their engagement in real life projects and professional associations will benefit the practice and research in the field. Academics in related programs, professional associations, and businesses should work together to create a culture of engagement. Faculty in the academic programs should stay active in related academic and professional organizations, seek projects, and involve their graduate students early in these projects. Academic and professional organizations such as AECT and ISPI should provide opportunities such as student projects, case competitions, and research and dissertation awards to encourage students pursuing advance

degrees in the field. Finally, business managers should welcome working with academics and graduate students alongside with their practitioners. Such collaboration can be facilitated via business and academic roundtables where business managers, academics, and practitioners equally participate in the exchange and testing of ideas. A business and academic roundtable would enable cross fertilization among key stakeholders of the HPT process and serve as a mentoring process for beginning practitioners and graduate students.

Compliance with Ethical Standards

Ethical approval: This article does not contain any studies with human participants performed by any of the authors.

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